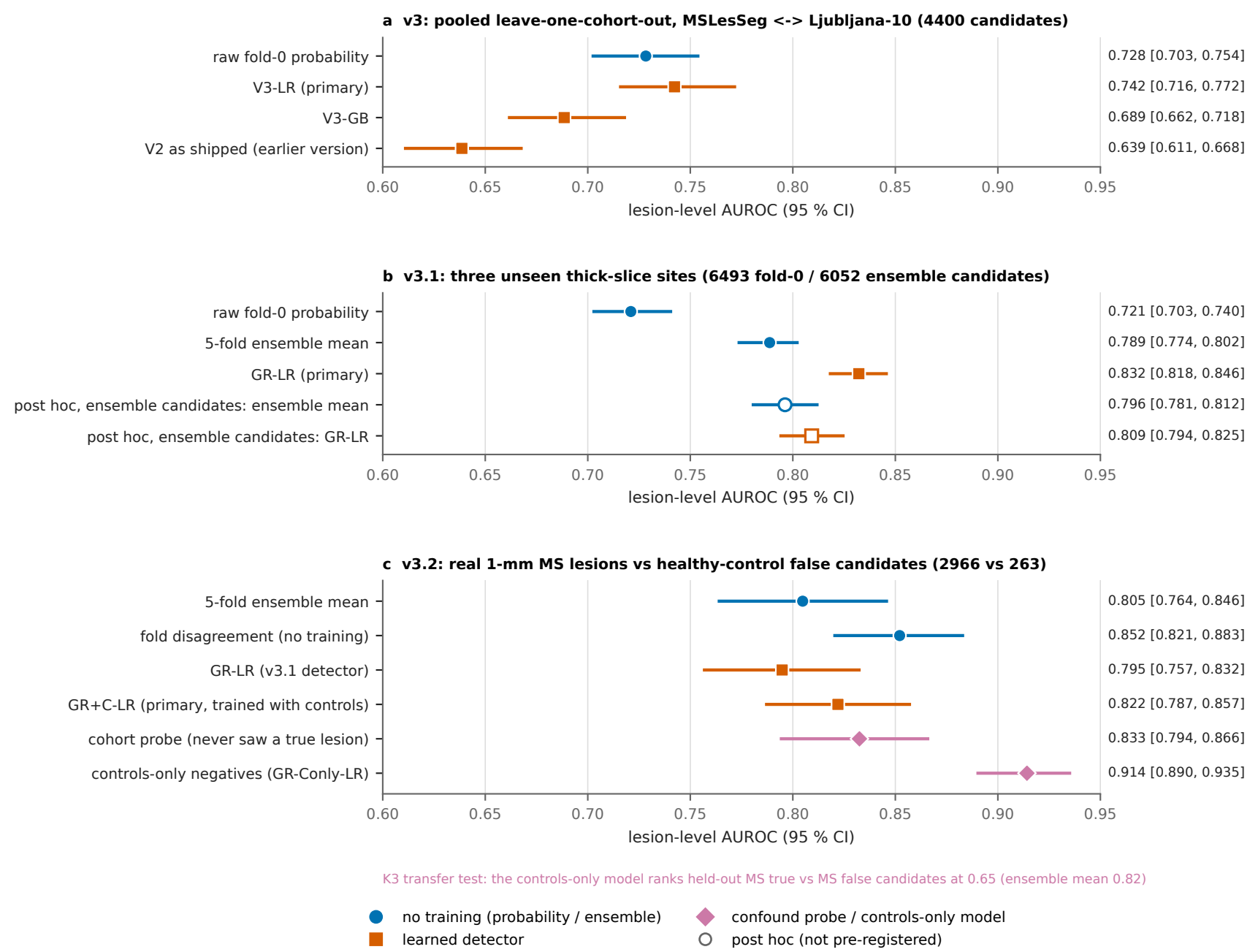
a v3: pooled leave-one-cohort-out, MSLesSeg <-> Ljubljana-10 (4400 candidates)
raw fold-0 probability
V3-LR (primary)
V3-GB
V2 as shipped (earlier version)
0.728 [0.703, 0.754]
0.742 [0.716, 0.772]
0.689 [0.662, 0.718]
0.639 [0.611, 0.668]
0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95
lesion-level AUROC (95 % CI)
b v3.1: three unseen thick-slice sites (6493 fold-0 / 6052 ensemble candidates)
raw fold-0 probability
5-fold ensemble mean
GR-LR (primary)
post hoc, ensemble candidates: ensemble mean
post hoc, ensemble candidates: GR-LR
0.721 [0.703, 0.740]
0.789 [0.774, 0.802]
0.832 [0.818, 0.846]
0.796 [0.781, 0.812]
0.809 [0.794, 0.825]
0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95
lesion-level AUROC (95 % CI)
c v3.2: real 1-mm MS lesions vs healthy-control false candidates (2966 vs 263)
5-fold ensemble mean
fold disagreement (no training)
GR-LR (v3.1 detector)
GR+C-LR (primary, trained with controls)
cohort probe (never saw a true lesion)
controls-only negatives (GR-Conly-LR)
0.805 [0.764, 0.846]
0.852 [0.821, 0.883]
0.795 [0.757, 0.832]
0.822 [0.787, 0.857]
0.833 [0.794, 0.866]
0.914 [0.890, 0.935]
0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95
lesion-level AUROC (95 % CI)
K3 transfer test: the controls-only model ranks held-out MS true vs MS false candidates at 0.65 (ensemble mean 0.82)
no training (probability / ensemble)
learned detector
confound probe / controls-only model
post hoc (not pre-registered)
Patient-clustered bootstrap 95 % CI (B = 2 000). Sources: lesion_detector_v3 / v31 / v32 evaluation and post-hoc records (2026-10-17, -21, -24).